

THIRUVENKADAPRASATH.S

192312016

12	Implement Iris Flower Classification using KNN
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CSV FILE

SepalLength,SepalWidth,PetalLength,PetalWidth,Species

5.1,3.5,1.4,0.2,Setosa

4.9,3.0,1.4,0.2,Setosa

4.7,3.2,1.3,0.2,Setosa

7.0,3.2,4.7,1.4,Versicolor

6.4,3.2,4.5,1.5,Versicolor

6.9,3.1,4.9,1.5,Versicolor

6.3,3.3,6.0,2.5,Virginica

5.8,2.7,5.1,1.9,Virginica

7.1,3.0,5.9,2.1,Virginica

PROGRAM

```
import numpy as np
```

```
import pandas as pd
```

```
class KNNScratch:
```

```
    def __init__(self, k=3):
```

```
        self.k = k
```

```
        self.X_train = None
```

```
        self.y_train = None
```

```
    def fit(self, X, y):
```

```

self.X_train = X
self.y_train = y

def _euclidean_distance(self, x1, x2):
    return np.sqrt(np.sum((x1 - x2) ** 2))

def _predict_sample(self, x):
    distances = [
        self._euclidean_distance(x, x_train) for x_train in self.X_train
    ]
    k_indices = np.argsort(distances)[: self.k]
    k_nearest_labels = [self.y_train[i] for i in k_indices]

    labels, counts = np.unique(k_nearest_labels, return_counts=True)
    return labels[np.argmax(counts)]

def predict(self, X):
    return np.array([self._predict_sample(x) for x in X])

df = pd.read_csv("iris_knn_data.csv")

X = df[
    ["SepalLength", "SepalWidth", "PetalLength", "PetalWidth"]
].values
y = df["Species"].values

model = KNNScratch(k=3)
model.fit(X, y)

```

```
predictions = model.predict(X)
```

```
accuracy = np.mean(predictions == y) * 100
```

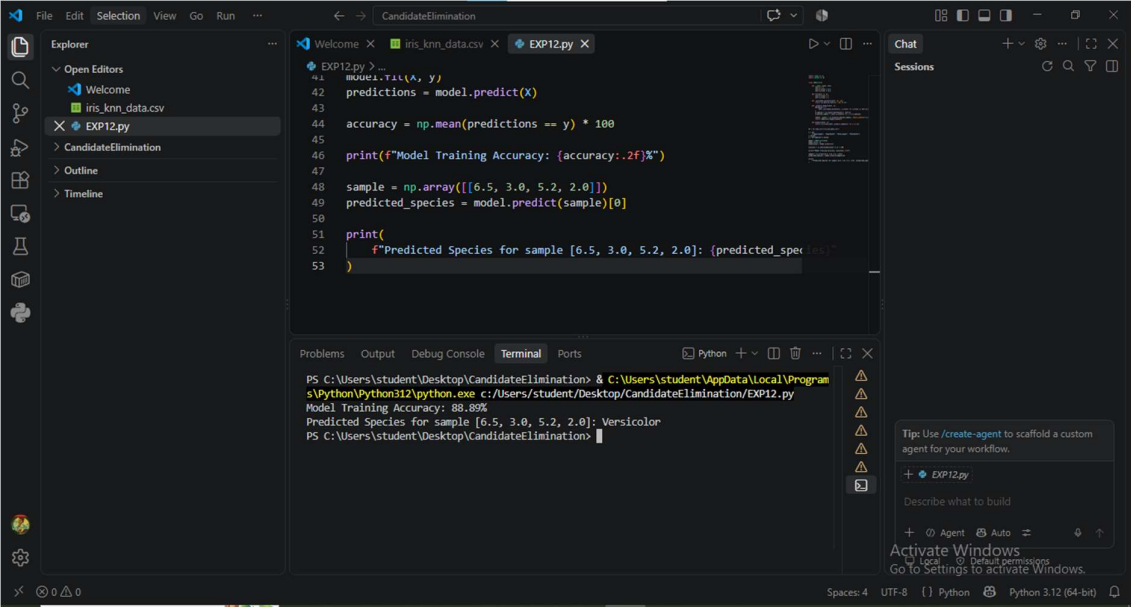
```
print(f"Model Training Accuracy: {accuracy:.2f}%")
```

```
sample = np.array([[6.5, 3.0, 5.2, 2.0]])
```

```
predicted_species = model.predict(sample)[0]
```

```
print(  
    f"Predicted Species for sample [6.5, 3.0, 5.2, 2.0]: {predicted_species}"  
)
```

OUTPUT



The screenshot shows a Visual Studio Code editor window with a file named 'CandidateElimination'. The editor contains a Python script that calculates the training accuracy of a model and predicts the species for a given sample. The terminal output shows the execution of the script, resulting in a training accuracy of 88.89% and a predicted species of 'Versicolor' for the sample [6.5, 3.0, 5.2, 2.0].

```
PS C:\Users\student\Desktop\CandidateElimination> & C:\Users\student\AppData\Local\Program  
Files\Python\Python312\python.exe c:/Users/student/Desktop/CandidateElimination/EXP12.py  
Model Training Accuracy: 88.89%  
Predicted Species for sample [6.5, 3.0, 5.2, 2.0]: Versicolor  
PS C:\Users\student\Desktop\CandidateElimination>
```